EmoSense: Multimodal Emotion & Wellness Assistant

COMPREHENSIVE PROJECT DOCUMENTATION & TECHNICAL ARCHITECTURE REPORT

Project Title:	EmoSense: Multimodal Emotion & Wellness Assistant	**Course / Code:**	MCA III CA550-Project (MO 2026)
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Execution Status:	**100% Fully Built & Deployed**	**Repository:**	github.com/D-dheeraj/Emosense

1. Executive Summary & Problem Statement

Mental health monitoring applications traditionally rely on single-modal inputs, such as self-reported text surveys or standalone facial expression analysis. These conventional approaches suffer from critical flaws, most notably **Emotional Masking** — a psychological phenomenon where individuals exhibit outward positive facial expressions (e.g., forced smiles) or select 'Happy' emojis while experiencing severe underlying distress, depression, or burnout.

EmoSense solves this challenge by implementing a privacy-first, multimodal agentic AI platform. The system captures three complementary signals during a user check-in: **Visual Emotion** (on-device PyTorch CNN), **Textual Sentiment** (VADER micro-journaling engine), and **Self-Reported Mood** (emoji picker). A **3-Way Congruence Agent** (Google Gemini API) triangulates these signals to detect emotional masking, a **Trend Scorer Agent** tracks baseline stability in SQLite, and a **Report Writer Agent** triggers real-time actionable micro-interventions (such as 4-7-8 guided breathing visualizers and 432Hz ambient audio relaxation).

Key Innovation: Privacy-preserving Edge AI face recognition running locally on Apple Silicon Mac M1 MPS GPU combined with Google Gemini multi-agent reasoning to catch hidden psychological distress without uploading raw video to external servers.

2. Complete Technology Stack & Specifications

Layer / Subsystem	Technology Used	Technical Specification & Purpose
Edge AI Hardware	Apple Silicon Mac M1 (MPS GPU)	On-device hardware acceleration via PyTorch Metal Performance Shaders (`torch.device('mps')`) for < 20ms face inference.
Deep Learning CNN	PyTorch 2.8.0 (`EmoNet`)	Custom 4-Block CNN trained on FER2013 dataset (35,887 images, 48x48 grayscale). Achieved **62.54% Validation Accuracy**.

Computer Vision	OpenCV 4.9.0 (`cv2`)	Decodes Base64 webcam frames, performs Haar Cascade face detection with auto center-crop fallback.
Text Sentiment Engine	VADER (`vaderSentiment`)	Rule-based lexicon analyzer returning compound sentiment score (-1.0 to +1.0) for micro-journaling entries.
Multi-Agent AI Core	Google Gemini API (`gemini-2.5-flash`)	Drives 3-Way Congruence Agent, Trend Scorer Agent, Report Writer Agent, and Universal Multilingual Guidance Chatbot.
Backend Framework	FastAPI + Uvicorn	Asynchronous REST API backend supporting `/api/checkin`, `/api/analyze-frame`, `/api/chat`, and `/api/history`.
Database & ORM	SQLite 3 + SQLAlchemy 2.0	Persistent storage for `UserCheckIn`, `EmotionLog`, and `InterventionLog` tables (`emosense.db`).
Frontend UI	HTML5 + Vanilla CSS + JS + Chart.js	Cyber-Health Dark Glassmorphic Dashboard with Tabbed Navigation, 4-7-8 Breathing visualizer, 432Hz Soundscape, and Chatbot Drawer.

3. Step-by-Step Implementation Methodology

The project was engineered systematically following a prototype-to-production lifecycle across 7 distinct phases:

Phase	Milestone Target	Key Work Completed & Deliverables
Phase 1	Environment Setup	Created virtual environment (`venv`), configured Apple Silicon PyTorch MPS device recognition, and initialized directory structure.
Phase 2	PyTorch CNN Model Training	Downloaded Kaggle FER2013 dataset (35k images). Wrote `app/ml/train_fer2013.py` and trained custom `EmoNet` CNN for 10 epochs on Mac M1 MPS GPU, achieving **62.54% accuracy**. Saved weights to `models_weights/fer2013_custom_cnn.pth`.
Phase 3	Edge Vision & NLP Engines	Built `face_detector.py` using OpenCV Base64 frame decoding with center-crop fallback, and `sentiment_analyzer.py` using VADER lexicon.
Phase 4	Multi-Agent AI Core	Implemented `congruence_agent.py` (Gemini API 3-Way Triangulation), `trend_scorer_agent.py` (ESI stability index), `report_writer_agent.py`, and `chatbot_agent.py` (Universal Multilingual Guidance Chatbot).
Phase 5	FastAPI Server & DB	Built `app/main.py` REST API routes (`/api/checkin`, `/api/analyze-frame`, `/api/chat`, `/api/history`) and SQLAlchemy SQLite models (`models.py`).
Phase 6	Glassmorphic Dashboard UI	Designed single-page dark Cyber-Health glassmorphic web dashboard with Tab Navigation, Webcam HUD, 4-7-8 Breathing visualizer, 432Hz Ambient Audio Player, and floating AI Chatbot drawer.
Phase 7	Git & GitHub Deployment	Configured `.gitignore` to protect datasets/keys, committed codebase, and pushed repository to **github.com/D-dheeraj/Emosense.git**.

4. System Modules & Core Functionalities

4.1 Multimodal Input Triangulation

• **Visual Signal**: Captures webcam video frame -> OpenCV crops face -> PyTorch MPS CNN classifies 7 facial emotions.
• **Textual Signal**: User enters micro-journal ('How was your day?') -> VADER computes sentiment score (-1.0 to +1.0).
• **Self-Reported Signal**: User taps an emoji (e.g. ■, ■, ■) to indicate subjective feeling.

4.2 Agentic AI Multimodal Triangulation Pipeline

• **Agent 1 (3-Way Congruence Agent)**: Sends signal triplet to Google Gemini API. Detects **Emotional Masking** (e.g., smiling face on camera + negative journal text like 'I feel helpless'). Computes Congruence Index (0-100%).
• **Agent 2 (Trend Scorer Agent)**: Queries historical check-in timeline from SQLite to calculate Emotional Stability Index (ESI) and prevent false alarms on isolated bad days.
• **Agent 3 (Report Writer & Micro-Intervention Agent)**: Generates plain-language wellness synthesis and assigns micro-interventions.
• **Agent 4 (Universal AI Companion Chatbot)**: Floating AI widget answering user questions in Hindi, Hinglish, or English about app usage, data privacy, tech stack, and breathing techniques.

4.3 Interactive Actionable Micro-Interventions Deck

• **4-7-8 Guided Breathing Visualizer**: Animated pulsating visual ring (Inhale 4s -> Hold 7s -> Exhale 8s) with timer countdown.
• **Ambient Soundscape Player**: Built-in 432Hz harmonic binaural relaxation sound generator to soothe nervous tension.
• **Data Privacy Center**: Interactive tab detailing on-device PyTorch MPS face detection and local SQLite database security.

5. Current Project Status & Milestone Verification

Project Component	Status	Empirical Verification Results
PyTorch Edge AI Model	**Trained & Verified**	Trained 10 Epochs on FER2013 dataset using Mac M1 MPS GPU. Saved at `models_weights/fer2013_custom_cnn.pth` with **62.54% Validation Accuracy**.
FastAPI Server & Endpoints	**Active & Verified**	All routes (`/api/checkin`, `/api/analyze-frame`, `/api/chat`, `/api/history`) tested and returning **HTTP 200 OK**.
Gemini Multi-Agent Core	**Active + Failsafe**	Google Gemini API integration verified with rule-based heuristic fallback for offline execution.
Glassmorphic Web Dashboard	**100% Deployed**	Live web dashboard running at **http://127.0.0.1:8000** with 4 Navigation Tabs, Floating AI Chatbot, and 4-7-8 Breathing Visualizer.
Git & GitHub Repository	**Committed & Pushed**	Codebase committed with clean `.gitignore` and pushed to **https://github.com/D-dheeraj/Emosense.git**.

6. Real-World Utility & Future Scope

1. **University & College Student Wellness Portals**: Pre-screens academic burnout, exam stress, and student depression prior to severe crises while preserving student biometric privacy.
2. **Corporate HR Employee Assistance Programs (EAPs)**: Integrated into IT company portals (TCS, Infosys, Accenture) to prevent tech worker burnout and prompt micro-breathing breaks.
3. **Tele-Psychiatry & Digital Therapeutics**: Provides psychiatrists with an empirical 30-day Emotional Stability Index graph between patient therapy sessions.
4. **Privacy-First Smart Home Health Kiosks**: Runs on edge AI hardware (Apple Silicon / Smart Mirrors) without sending raw camera video to cloud servers.

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